EXPERIMENT No 5:

WIRED REMOTE CONTROLLED CAR

Student Guide

Prepared by BLIS GLOBAL Robotics Department

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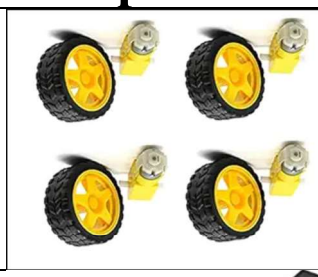
A wired remote-controlled robot car using electricity only is a basic robotic vehicle that is operated through a physical wire connection between the control unit (remote) and the robot car. The robot is powered by electricity, typically from a battery, and the user controls its movements (e.g., forward, backward, left, right) by sending electrical signals via the wired remote.

Components Needed:

S/No	Component Name	Components Image
	4 DC motors (for the wheels)	
	2 switches (SPDT or DPDT switches for directional control)	
	Battery (for powering the motors)	

	Robot Frame		
	Wires for connections		

Basic Principle:

To move forward, backward, left, and right, you will manually control the direction of the current flowing into the motors. This will determine whether the motors spin clockwise or counterclockwise, thus moving the robot forward, backward, or making turns.

Step-by-Step Process:

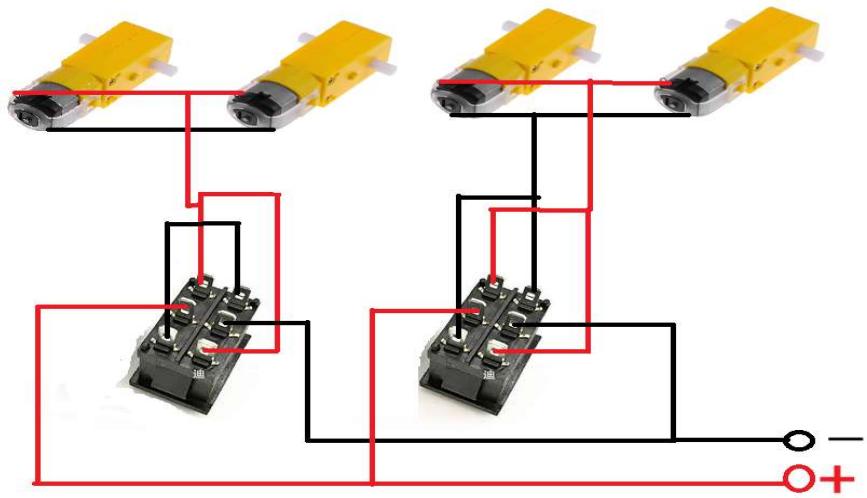
1. Wiring the Motors:

- Each motor should be connected to its respective wheel.
- You need to be able to reverse the current in the motors to change the direction of rotation (for forward and backward motion).

2. Using DPDT Switches:

- A DPDT switch (Double Pole Double Throw) allows you to easily reverse the polarity of the motors to control the direction.
- **You'll need 2 DPDT switches for direction control:**
  - One switch will control the left and right motors.
  - The other switch will control the front and back motors.

Wiring Diagram for Each Motor Using DPDT:



- Connect the positive terminal of the battery to one end of the switch.
- Connect the negative terminal of the battery to the opposite side of the switch.
- The output from the DPDT switch will go to the motor terminals. Flipping the switch will reverse the polarity and thus the motor direction.

3. Controlling Forward and Backward:

- To move forward, both left and right motors should rotate in the same direction.
- To move backward, flip both DPDT switches to reverse the current direction on both sides.
- Make sure both switches are in sync (both flipped forward for forward movement or backward for reverse).

4. Controlling Left and Right Turns:

- To turn left, you need to stop or slow down the left-side motors while keeping the right-side motors running forward.
- Similarly, to turn right, stop or slow down the right-side motors while the left-side motors keep running forward.
- You can use SPST switches (Single Pole Single Throw) to simply turn off the motors on the side you want to stop.

5. Combining Switches for Full Control:

- Use 2 DPDT switches for the polarity control (for forward and backward control of left and right-side motors).
- Use SPST switches for turning (one for each motor or side of motors).

6. Switch Setup:

- Switch 1: Controls the front two motors (left and right).

- Switch 2: Controls the back two motors (left and right).
- Switch 1 and 2: Can be used to disable either the left side or right-side motors for turning.

7. Power Supply:

- Ensure your battery has sufficient voltage and current to power all 4 motors.
- Connect the battery to the common terminals of the switches, ensuring proper connections.

Summary of Movements:

- Forward: Flip both DPDT switches to the "forward" position.
- Backward: Flip both DPDT switches to the "reverse" position.
- Turn Left: Use SPST switches to stop the left-side motors, keeping the right motors running forward.
- Turn Right: Use SPST switches to stop the right-side motors, keeping the left motors running forward.

This setup will allow you to control your robot car manually without an Arduino or motor shield by just using switches to control motor direction and power.